

David Tench

High-Performance Algorithms Engineer

CONTACT INFORMATION Email: [dtench \[at\] pm \[dot\] me](mailto:dtench[at]pm[dot]me) Github: <https://github.com/tenchd>
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SKILLS I design and build high-performance systems for graphs and databases that use randomized algorithms to massively scale up and scale out applications in domains like bioinformatics and databases.

PROFESSIONAL EXPERIENCE **Lawrence Berkeley National Lab**, Berkeley, CA
Grace Hopper Postdoctoral Fellow **June 2023 – June 2026**
Rutgers University, New Brunswick, NJ
CRA Computing Innovation Postdoctoral Fellow **Aug 2021 – June 2023**

SKILLS

- Design and analysis of massive-scale algorithms and data structures
- Expertise in graph algorithms/systems, out-of-core computation (distributed, memory-hierarchy-optimized), randomized algorithms, parallelism
- Writing high-performance Rust & C++ code. Ex: https://github.com/tenchd/spec_spars
- Profiling and diagnosing performance of high-performance implementations
- Leading a high-performance algorithms coding team (17 students)
- Technical writing: 13 peer-reviewed papers, 4 grant applications
- Technical presentation: 18 conference or workshop talks

SELECTED PROJECTS **Linear Sketching for Large-Scale Graph Processing**
Team Lead, Algorithm Designer

- Noticed 1) that existing graph processing systems fail to scale to massive, dense, dynamic graphs and 2) algorithms in theory literature (called **graph sketches**) that might help, but had never been implemented due to high computational and data movement costs.
- Designed new randomized graph sketching algorithms that simplify and reduce the computational cost of the SOTA, and optimize for out-of-core (distributed, SSD) computation.
- Recruited and led 8-student team to implement sketching algorithms through GraphZeppelin, a high-performance multi-core graph processing system. C++14, OpenMP, MPI.
- 1st author paper appeared at SIGMOD, world's top-ranked database CS research conference. Followup last-author paper to appear at SIGMOD in 2027.
- 1st author paper, a distributed version (increasing throughput by more than 10x), appeared at ALENEX, a top-ranked algorithmic engineering CS research conference.
- Papers currently in submission to top-ranked conferences expand the above ideas to GPU acceleration, fast dynamic connectivity queries, and hybrid sketching algorithms. See project repos at <https://github.com/GraphStreamingProject>.

Randomized Graph Algorithms for Compacting Memory Management

Algorithm Designer/Engineer

- Designed real-time randomized matching graph algorithms used to reduce C++ application memory footprint by compacting objects without changing virtual addresses.
- Experimentally verified runtime and solution quality of novel algorithms.
- Co-1st author paper appeared at PLDI, a top-ranked operating systems conference.
- Mesh, our C++ compacting memory manager: <https://github.com/plasma-umass/Mesh>

EDUCATION **University of Massachusetts Amherst**, Amherst, MA
Ph.D., Computer Science and M.S., Computer Science **Aug 2014 – Aug 2020**
Dissertation title: "Algorithms for Massive, Changing, and Otherwise Inconvenient Graphs"
Lehigh University, Bethlehem, PA
B.S., Mathematics **Aug 2009 – May 2013**